

Executive Summary: FragmentedID Resolution

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Fragmented identity data creates direct operational and financial risk when the same person appears under name, address, or alias variations. In healthcare, this can contribute to duplicate patient records and care coordination failures. In financial services and credit operations, it can slow onboarding, increase manual review, and weaken synthetic identity detection. The LinkedID application addresses this by identifying likely duplicates even when records differ in realistic ways such as nickname use, surname expansion, typos, and address formatting differences.

This project was trained and evaluated on the Hasso Plattner Institutes (HPI) snap shot of the North Carolina Voter Registration dataset as a public, privacy-safe proxy for real-world identity resolution. The dataset includes 14,183 prepared records with labeled duplicate (9,891) and non-duplicate pairs (98,142). While this is not production banking or healthcare data, it provides a credible benchmark for testing entity matching methods under noisy, fragmented identity conditions in these environments.

Results show clear business value beyond model novelty. The final deployed model, a hard-weighted Siamese BiLSTM, achieved strong test performance with F1 of 0.9935 and PR-AUC of 0.9998. On the blocked test set, it produced only 1 false positive and 2 false negatives across 2291 evaluated pairs. Compared with a strong Term Frequency-Inverse Document Frequency (TF-IDF) baseline, the deep model delivered the largest gains on difficult cases that typically drive analyst workload and customer friction. Hard-case F1 improved from 0.71 to 0.96 in the tuned Siamese model, indicating materially better behavior on ambiguous identity collisions.

Operationally, the system is built for decision support, not just offline scoring. The deployed workflow includes confidence scores, threshold tuning, and a human review queue with persisted reviewer decisions. This enables risk-based policy design: organizations can auto-merge very high-confidence matches, route uncertain pairs to analysts, and tune thresholds to align with business tolerance for false merges versus missed duplicates. In practice, that means lower manual review burden, faster resolution time, and higher-quality identity graphs for downstream decisions such as Know Your Customers (KYC), fraud screening, and customer record consolidation.

A key practicality finding is that shared blocking significantly reduces scoring volume while preserving duplicate coverage. In our tests, candidate pairs dropped from 2291 to 303, an 86.8 percent reduction, with a positive pass rate retained at 100 percent. This supports scalability of the approach and lowers compute cost per decision. However, full enterprise latency and throughput benchmarking under production traffic has not yet been completed, so current results should be treated as pilot-ready evidence rather than proof of final Service Learning Agreement (SLA) compliance.

Implications for stakeholders are straightforward. For healthcare operators, better linkage can reduce duplicate chart risk and manual reconciliation work. For financial institutions, better identity resolution can shorten onboarding cycles, reduce avoidable KYC rework, and strengthen synthetic fraud defenses. For credit bureaus, improved consolidation accuracy supports cleaner credit files and more reliable dispute handling.

Recommendation: Proceed with phased deployment. Start with human-in-the-loop operation, monitor manual review reduction and time to resolution as primary business KPIs, and then expand automation after validating performance on in-domain production-like data and confirming latency, throughput, and governance controls.